

Vinh Ly

Palo Alto, CA | vily437@outlook.com | (318) 202-8037
<https://www.linkedin.com/in/vinh-ly-14b56b50>

Summary

Software Engineer with **18 years** building and owning distributed systems at scale. Led infrastructure reliability for **YouTube** playback serving billions of daily requests and **Google** search serving millions of users. Designed real-time data pipelines processing **terabytes per hour** across **50+ microservices**. Expert in **C++**, **Go**, **Kubernetes**, and **GCP**. Managed teams of 8 engineers driving platform stability and feature delivery. Seeking to apply deep infrastructure and systems expertise to data platform challenges. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages: C++, Go, Python, Java, Bash, TypeScript, JavaScript, HTML, CSS

Infrastructure: Kubernetes, Docker, Terraform, GCP, BigQuery, Pub/Sub

Data: Kafka, Redis, PostgreSQL, Dataflow, Real-time streaming pipelines

Observability: Prometheus, Grafana, Splunk, On-call rotation, SLO/SLA management

Practices: CI/CD pipelines, Chaos engineering, Distributed systems, End-to-end ownership, AWS

Work Experience

Staff Software Engineer

YouTube | 02/2025 - Present | Remote

- Designed tooling for automated remediation and self-healing infrastructure reducing mean time to recovery from hours to minutes.
- Owned core playback and recommendation infrastructure serving billions of video requests daily with 99.99% uptime requirements.
- Implemented chaos engineering practices and game day exercises that reduced production incidents by 35% quarter-over-quarter.
- Collaborated with video encoding, CDN, and frontend teams to optimize end-to-end latency across the playback pipeline.
- Participated in on-call rotation for critical infrastructure covering 24/7 incident response and escalations.

Staff Software Engineer

Whatnot | 05/2024 - 02/2025 | Remote

- Built platform reliability infrastructure for a livestream commerce marketplace handling thousands of concurrent streams during peak sales events.
- Architected migration of core services from monolithic architecture to **Kubernetes**-based microservices with zero downtime deployments.
- Designed **real-time data pipelines** processing terabytes of event data hourly for analytics and ML feature generation.
- Established observability standards using **Prometheus** and **Grafana** dashboards tracking SLOs for critical user journeys.
- Implemented **Terraform** modules standardizing infrastructure provisioning across staging, canary, and production environments.

Staff Software Engineer / Manager

Google | 03/2015 - 04/2024 | Remote

- Led a team of 8 engineers building distributed systems infrastructure for **Google Search** serving millions of queries per second across global data centers.
- Redesigned caching architecture reducing backend query load by 40% and improving cache hit rates from 65% to 92%.
- Owned reliability and performance for search serving stack, achieving sub-100ms p99 latency targets for core ranking pipelines.
- Mentored engineers on systems design, conducted performance reviews, and drove technical roadmap alignment with product leadership.
- Led migration of search infrastructure to containerized workloads on **Kubernetes**, decommissioning legacy **Borg** deployments.

Software Engineer

Google | 02/2011 - 02/2015 | Remote

- Developed backend services in **C++** for search ranking pipeline handling millions of documents per second during index updates.
- Optimized database query patterns and implemented **Redis** caching layers that improved response times by 45% for **热门** queries.
- Debugged kernel-level issues in distributed systems requiring deep analysis of network latency and memory allocation patterns.
- Built **CI/CD pipelines** automating testing and deployment of search infrastructure components across multiple environments.

Software Design Engineer II

Microsoft | 10/2007 - 12/2011 | Remote

- Developed monitoring and alerting systems in **C++** that detected storage cluster failures before user impact.
- Designed and implemented core components for **Windows Azure** storage services handling petabyte-scale data operations.
- Built distributed consensus protocols for data replication and consistency guarantees across geographically distributed nodes.
- Contributed to **Linux**-based infrastructure supporting Azure cloud platform services and customer deployments.

Software Design Engineer Intern

Microsoft | 06/2006 - 08/2006 | Remote

- Developed tooling for automated testing of **distributed systems** components in support of Azure platform engineering team.

Education

B.S. Honors Degrees

Virginia Tech | 2004 - 2007